

Disagreement-Aware Probabilistic U-Net: Making Predictive Diversity Land Where Annotators Actually Disagree

Amit Heshes ([**TODO: ID**])

Idan Smoller ([**TODO: ID**])

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Title page details

- **Selected paper:** S. Kohl et al., *A Probabilistic U-Net for Segmentation of Ambiguous Images*, NeurIPS 2018.
- **Proposed extension:** supervise *where* human annotators disagree, with (i) a head that predicts the inter-rater disagreement map from the image alone and (ii) a loss that pushes the model’s own sample diversity into those same pixels.
- **Headline outcome:** the head clearly works (+0.20 correlation over a trivial control); the alignment loss gives a small but consistent gain (+0.026) at the best generalized energy distance of our arms. A fourth arm that replaces the paper’s objective with direct distribution matching wins on Dice but collapses to zero sample diversity — a negative result we report and explain.

1 Introduction

The problem, for a reader new to the task.

- Segmentation labels every pixel (here: is this pixel part of a lung nodule?), and training normally assumes one correct mask per image. In medical imaging that assumption is false: experts outline the same lesion differently, and on LIDC-IDRI they sometimes disagree about whether a lesion is there at all.
- A standard U-Net trained on one expert’s mask, or on their average, emits a single confident mask and destroys that information.

What the paper does.

- The Probabilistic U-Net combines a U-Net with a conditional VAE. A prior network maps the image x to a distribution $p(z | x)$ over a low-dimensional latent (6-D here); every draw of z decodes to a different plausible segmentation — a *coherent whole mask*. That is the paper’s central design choice: it argues explicitly against pixel-wise probability maps, because they cannot express which pixels vary *together*.
- Training minimises the negative ELBO. A posterior $q(z | x, Y_g)$ additionally sees *one randomly chosen* expert mask Y_g :

$$L_{\text{PU-Net}} = \underbrace{\mathbb{E}_{z \sim q(z|x, Y_g)} [-\log p(Y_g | x, z)]}_{\text{reconstruction (cross-entropy, summed over pixels)}} + \beta \underbrace{D_{\text{KL}}(q(z | x, Y_g) \parallel p(z | x))}_{\text{pulls the prior towards the posterior}} .$$

The balance between the two terms is not cosmetic: Section 5 describes how getting the reconstruction’s reduction wrong collapses the posterior onto the prior and destroys sampling entirely.

- Sampling z repeatedly therefore yields a *distribution* of segmentations, scored against the four expert masks with the **generalized energy distance**. Writing S, S' for independent model samples and Y, Y' for independent grader masks,

$$D_{\text{GED}}^2 = 2 \mathbb{E}[d(S, Y)] - \underbrace{\mathbb{E}[d(S, S')]}_{\text{sample diversity}} - \mathbb{E}[d(Y, Y')], \quad d = 1 - \text{IoU}.$$

Why it matters, and why we chose it.

- In nodule screening the disputed cases are the clinically interesting ones: a model that says “experts would argue *here*” is more useful as a second reader than one that draws a confident wrong boundary. LIDC ships **four** expert masks per image, so disagreement is directly measurable — rare among segmentation datasets.

- **The gap — read the middle term of D_{GED}^2 twice.** Every term of GED is a *scalar*: $E[d(S, S')]$ says how *much* the model’s samples vary and $E[d(Y, Y')]$ how much the graders vary, but nothing in it says *where*. A model could put the right *amount* of variability in the wrong *places* and score just as well. The “where” question needs the two *maps* behind those scalars — U for the samples, D^{GT} for the graders (Section 2) — and comparing them pixel by pixel is precisely what we add.

2 Proposed extension

Three quantities, kept distinct throughout the report.

- D^{GT} — *human* disagreement, computed from the four masks. With $P_i = \frac{1}{G} \sum_g Y_i^{(g)}$ the per-pixel mean over $G = 4$ graders, D^{GT} is its normalized binary entropy

$$D_i^{\text{GT}} = -\frac{P_i \log(P_i + \epsilon) + (1 - P_i) \log(1 - P_i + \epsilon)}{\log 2} \in [0, 1],$$

so $D^{\text{GT}} = 0$ where all four agree and 1 where they split two-two.

- $\hat{D}$ — the *prediction* of D^{GT} made by our new head.
- U — *model* uncertainty. Draw K latents from the prior, decode each to a foreground-probability map $Q^{(k)}$, and average them into $\bar{Q}_i = \frac{1}{K} \sum_k Q_i^{(k)}$. Then U is the *same functional* as D^{GT} , with the K samples playing the role of the G graders:

$$U_i = -\frac{\bar{Q}_i \log(\bar{Q}_i + \epsilon) + (1 - \bar{Q}_i) \log(1 - \bar{Q}_i + \epsilon)}{\log 2}.$$

Using one functional on both sides is what makes “ $U \approx D^{\text{GT}}$ ” a claim about *where* uncertainty sits rather than a comparison of two different units.

What we add (two pieces, both minimal on purpose).

- **(B) Disagreement head.** One 1×1 convolution + sigmoid on the last U-Net feature map, tapped *before* z is injected, so $\hat{D}$ is a function of the image alone. Trained with $L_D = \text{MSE}(\hat{D}, D^{\text{GT}})$.
- **(C) Alignment loss.** $L_{\text{align}} = \frac{1}{HW} \sum_i |U_i - D_i^{\text{GT}}|$, penalising predictive diversity that lands anywhere other than where the graders differ. Full objective $L = L_{\text{PU-Net}} + \lambda_D L_D + \lambda_A L_{\text{align}}$; in Figure ?? everything in the dashed box is new.

Where our addition sits: a marginal readout on a joint model.

- D^{GT} and U are per-pixel *marginals* — projections of the joint distribution over masks. We do not swap one object for the other: the CVAE still owns the joint (the KL keeps the prior wide and the samples coherent), and L_{align} only says *where* the marginal variance should sit. U is computed from the K decoded samples, so it is a *readout* of that distribution, not a parallel pixel-wise model.
- Four masks pin the per-pixel Bernoulli down exactly — with $G = 4$, D^{GT} can only take the values $\{0, 0.81, 1\}$ — but say very little about the joint. A marginal constraint is roughly all the labels can support.
- **Caveat we state rather than gloss:** $\hat{D}$ is a pixel-wise predictor, a 1×1 conv on image features with no sample involved. It is an auxiliary output *alongside* the sample-based segmentation, never a substitute for it.

Hypothesis, as two testable claims.

- **Q1.** Where annotators will disagree is predictable from the image alone: $\hat{D} \approx D^{\text{GT}}$, beating a trivial “just outline the lesion” control.
- **Q2.** Adding L_{align} raises the agreement between U and D^{GT} over the baseline, without damaging segmentation quality or GED.

Alternatives we considered.

- **Condition the prior on disagreement, $p(z | x, \hat{D})$.** Rejected: $\hat{D}$ is itself a deterministic function of x , so it adds no information and confounds the ablation.
- **A dedicated multi-rater architecture.** MRNet [?] models rater (dis)agreement with an expertise-aware module and extra branches. Rejected for this project: with a single 1×1 convolution, any gain is attributable to the *supervision signal* rather than to added capacity.

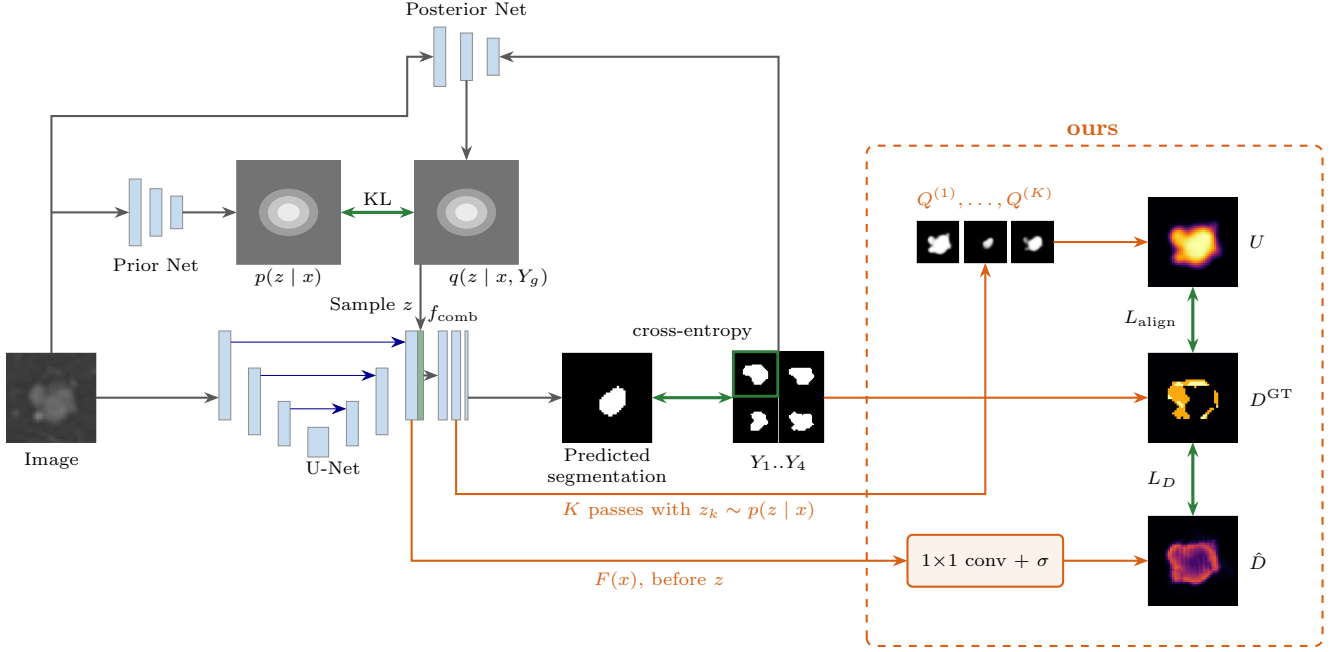


Figure 1: **The disagreement-aware Probabilistic U-Net (training).** Outside the dashed box, the original model: the posterior sees the image and *one* randomly drawn grader Y_g (green outline), and the sampled z (green block, 6 channels) is concatenated to the last U-Net activation (32 channels) and the three 1×1 convolutions of f_{comb} combine them; block width reads as channel count. Inside the box, our extension. Two things the drawing encodes: $F(x)$ is tapped *before* z , so $\hat{D}$ depends on the image alone; and the K passes behind U draw fresh z_k from the *prior*. **Every thumbnail is real**, from test crop LIDC-IDRI-0217 decoded by our trained full model and zoomed to the nodule; D^{GT} , U and $\hat{D}$ share one $[0, 1]$ colour scale, so they can be compared directly. The three $Q^{(k)}$ shown are non-empty draws — 8 of the 16 samples for this crop are empty, though all four graders segmented the nodule.

- **Per-rater heads / confusion matrices** [?]. Rejected: the LIDC release does not identify which radiologist produced which mask and the four raters differ across scans, so a per-rater parameterisation has nothing stable to learn.
- **Replace the ELBO with direct distribution matching.** Treat the four masks as an empirical distribution and train the K samples to match it with a two-way soft-Dice loss — close in spirit to MoSE [?], which matches a segmentation distribution to the annotation set with a Wasserstein-like loss on LIDC. Rather than only discussing it, **we implemented and ran it as a fourth arm (D).**

3 Methodology

Model.

- Faithful PyTorch re-implementation of the Probabilistic U-Net (the official repo is old TensorFlow and does not run on current Colab): U-Net backbone with base 32 channels and 4 down/up-samplings, prior encoder, posterior encoder, 6-D latent, and f_{comb} -style injection of z into the final 1×1 convolution stack.
- Our additions live in a subclass, so the baseline stays runnable for a controlled comparison.
- **A bug in our own re-implementation, worth reporting.** The final 1×1 stack initially had no nonlinearity between its convolutions, so it collapsed to a single affine map; since the injected z is spatially constant, every sample became a level set of one function and z could only globally grow or shrink one mask. Measured effect: **100% of test images had strictly nested samples.** Adding the missing activation dropped that to 1–3%, and is what let any diversity-shaping loss work at all.

Data.

- LIDC-IDRI 2D crops, DeepMind’s preprocessed release (CC BY 3.0) published with the Hierarchical Probabilistic U-Net [?]; preprocessing follows the original paper’s Appendix H.1.
- Splits: 8843 train / 1993 val / 1980 test images (530 / 111 / 103 patients), 180×180 8-bit grayscale, four masks each. We take a random 128×128 crop, scale images to $[0, 1]$ and binarise masks. These counts are $\sim 0.4\%$ below the 8882 / 1996 / 1992 quoted in the paper, because the authors cleaned the data after publication and the 2018 snapshot was never released; we therefore cite the HPU-Net release and do not claim a byte-identical reproduction.
- **Roughly 65% of training crops contain at least one empty mask** — graders disagree about lesion *existence*, not only about the boundary. This is signal, not corruption, and it is why the *empty-mask convention* defined in Section 5 matters.

Training.

- Loss: summed cross-entropy reconstruction + β KL ($\beta = 1$, 6-D latent, per Appendix H.1), plus $\lambda_D L_D$ and $\lambda_A L_{\text{align}}$ for arms B and C.
- Adam, lr 10^{-4} decayed to 10^{-6} in 5 steps, batch 32, weight decay 10^{-5} , gradient-norm clipping at 100, $K = 4$ samples per step, 16 samples per image at test time; augmentation is the random translated crop only.
- **100k steps per arm** (the paper uses 240k), one GPU, ≈ 3.7 h for A/B, ≈ 6.6 h for C, ≈ 6.0 h for D.
- $\lambda_D = 0.01$ and $\lambda_A = 10^{-5}$; both are load-bearing. L_D is summed over pixels, so $\lambda_D = 0.5$ makes it $\sim 3\times$ the reconstruction term; and $\lambda_A = 10^{-4}$ diverges within 2k steps because inflating the prior variance is the cheapest way to raise diversity (Section 5).
- Tools: Python, PyTorch, NumPy, Matplotlib. The submitted code runs in Google Colab: a self-contained notebook executes the whole pipeline end-to-end in a reduced “smoke” mode, and the same code paths, run at full length on one GPU, produced every number below.

Ablation design (one variable at a time).

Arm	Disagreement head	Alignment loss	Objective
A – baseline PU-Net	no	no	$L_{\text{PU-Net}}$
B – + head	yes	no	$+\lambda_D L_D$
C – full model	yes	yes	$+\lambda_D L_D + \lambda_A L_{\text{align}}$
D – distribution matching	yes	no	$\lambda_{\text{dist}} L_{\text{dist}} + \lambda_{\text{con}} L_{\text{con}} + \lambda_D L_D$

- **Arm D in detail.** The ELBO is dropped entirely (no random-grader cross-entropy, no posterior, no KL). Instead K samples are drawn from the prior *with gradients*, and a $K \times G$ matrix of soft-Dice distances to the four masks is reduced two ways: *coverage* (every human mask must be matched by some sample and vice versa, via a soft minimum) or *kernel* (a two-way soft kernel likelihood). L_{con} is a soft Dice between the mean sample and the mean human mask. Both modes are reported.

4 Results and analysis

Metrics, and what each is for.

- **Dice / IoU** — sanity check that the extension does not damage segmentation quality, computed on the deterministic prediction against each of the four masks. *Not* the headline claim.
- **Generalized energy distance (GED)**, the paper’s own metric, defined in Section 1: our 16 samples against the four human masks, with $d = 1 - \text{IoU}$. We evaluate it under the **empty-mask convention** of the paper’s Appendix B — $d \equiv 0$ when both masks are empty, so agreeing that there is no lesion is rewarded rather than punished. We apply the same convention to Dice/IoU by excluding empty-vs-empty pairs instead of scoring them 0; without it the 65% of crops with an empty mask deflate every model and hide real differences.
- $\text{corr}(U, D^{\text{GT}})$ — does the model become uncertain in the same places humans disagree? **This is the primary result.**
- **Trivial control** — disagreement in LIDC sits largely on lesion boundaries, so a map that merely marks the boundary already correlates with D^{GT} . Ours is the 5 px *boundary band* (dilation – erosion) of **the model’s own predicted mask** — roughly 2 px inside and 2 px outside its edge. It uses no grader information whatever, so unlike a band drawn around the ground truth it is available at inference. $\hat{D}$ must beat it to have learned anything about *raters* rather than about outlines.

- **Diversity diagnostics** — the subtracted $E[d(S, S')]$ term of GED, reported on its own, and the fraction of images whose samples are strictly nested. Together they separate “diverse in the wrong place” from “not diverse at all”.

Main table. Held-out test split, 1980 images, 16 samples per image, all arms read from the matched 100k-step checkpoint.

Arm	Dice $\uparrow$	IoU $\uparrow$	GED $\downarrow$	$\text{corr}(U, D^{\text{GT}}) \uparrow$	$\text{corr}(\hat{D}, D^{\text{GT}}) \uparrow$	control	$E[d(S, S')]$	nested $\downarrow$
A – baseline	0.355	0.280	0.394	0.483	—	0.417	0.686	1%
B – + head	0.308	0.241	0.403	0.492	0.609	0.451	0.687	3%
C – full	0.340	0.267	0.388	0.509	0.627	0.425	0.692	1%
D – coverage	<i>0.438</i>	<i>0.369</i>	0.714	0.120	0.565	0.507	0.024	99.5%
D – kernel	<i>0.405</i>	<i>0.337</i>	0.933	0.103	0.569	0.426	0.000	100%

control is the trivial “boundary band of your own prediction” baseline for the same model, so $\text{corr}(\hat{D}, D^{\text{GT}})$ should be read against the number beside it. Dice for arm D is italicised because it is not comparable — see below.

Q1 — can the model predict where annotators will disagree? Yes.

- The head reaches $\text{corr}(\hat{D}, D^{\text{GT}}) = 0.627$ against its own boundary-band control of 0.425: +0.20. It has learned something about rater behaviour, not merely where lesion edges are. This is the project’s clearest positive result and it holds in every arm with a head.

Q2 — does the alignment loss move uncertainty to the right place? Weakly yes.

- $\text{corr}(U, D^{\text{GT}})$ rises $0.483 \rightarrow 0.509$ from A to C, and C simultaneously attains the *best* GED of the three ELBO arms (0.388), for a 4% relative Dice cost.
- The gain is modest, and we read that as *structural* rather than a tuning failure: L_{align} constrains a marginal of a joint object, so two sample distributions with identical marginals are invisible to it. It can relocate variance; it cannot reshape modes.
- It only became visible at all after the `fcomb` fix of Section 3: with 100% nested samples U is a single ring around one mask, and no loss can reshape it.

Q3 — and arm D: direct distribution matching wins Dice and loses everything else.

- Arm D posts the best Dice by a wide margin (0.438 vs. 0.355) and the worst GED by a wide margin (0.714 / 0.933 vs. 0.388).
- **The mechanism is in one column:** $E[d(S, S')]$ is **0.024** (coverage) and **0.0004** (kernel), against 0.69 for arms A–C, with **99–100% of test images nested**. All 16 samples are effectively the same mask — the model has become deterministic, and the subtracted diversity term of D_{GED}^2 vanishes, which is exactly why GED nearly doubles while Dice improves.
- **Why.** The two-way soft-Dice objective plus the consensus term is minimised by placing *every* sample on the single mask closest to all four annotations. Nothing in it rewards spread. In the original ELBO the KL term keeps the prior wide; deleting the CVAE machinery deleted the only pressure holding the samples apart.
- The Dice win is therefore circular: arm D is trained on a soft-Dice criterion and then evaluated by Dice on what is now a single consensus mask. It is a better *point* estimate and a far worse model of ambiguity, which is the thing the paper set out to capture.
- **Arm D is the mirror image of our extension.** It tries to obtain the joint *from the loss* — matching whole masks set-to-set — after deleting the machinery that maintained it, and ends up with no joint at all. We keep the CVAE’s joint and constrain only its marginal.
- **A metric lesson worth stating.** Arm D also achieves the *lowest* disagreement MAE (0.0125 vs. 0.0176 for C), because D^{GT} is near zero almost everywhere and predicting “no uncertainty anywhere” scores well on a mean absolute error. Correlation exposes this immediately; MAE does not. We report both and rely on correlation.

Failure cases and challenges encountered.

- **Posterior collapse (first full run, discarded).** Our reconstruction term averaged cross-entropy over pixels while Kohl et al. sum it; at 128×128 that under-weights reconstruction $\sim 16000\times$ against β KL, so $\beta = 1$ behaved like $\beta \approx 16000$. The KL hit exactly 0 by 30k steps, sample diversity fell from 0.45 to 0.006, and val GED *rose* $0.34 \rightarrow 0.61$ while Dice improved. Fixed by summing the cross-entropy.

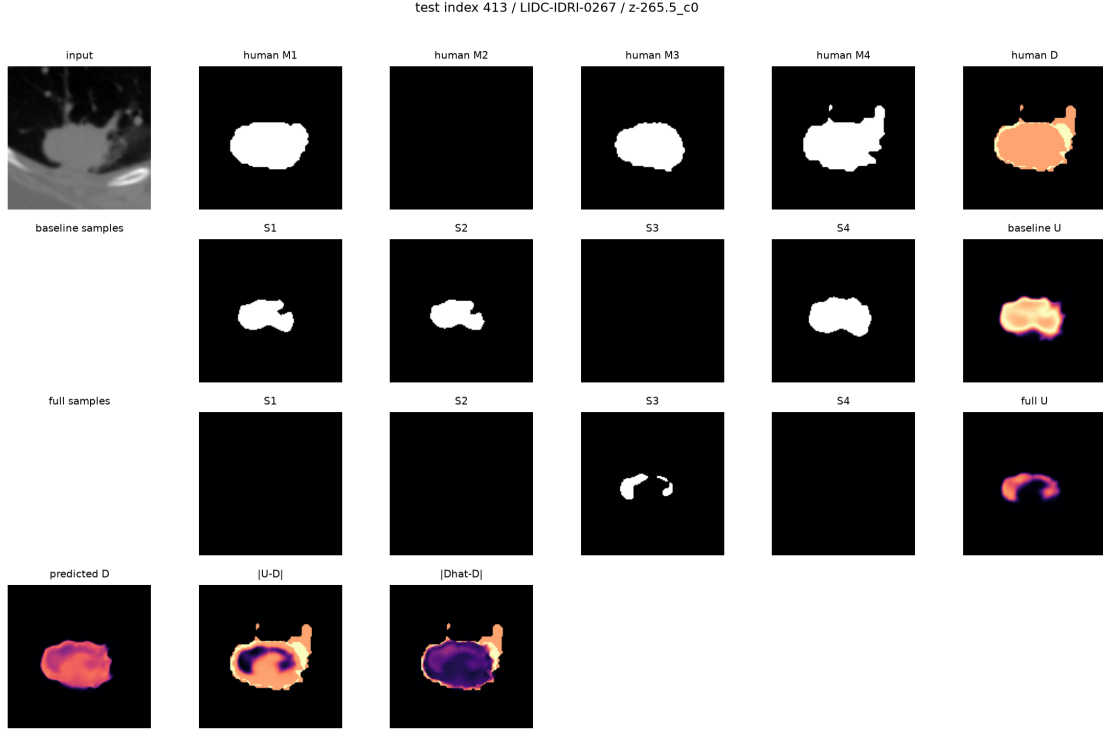


Figure 2: **A case where the graders disagree about whether a lesion exists at all** (test index 413, LIDC-IDRI-0267). Row 1: the CT crop, the four expert masks — note that **M2 is empty** — and D^{GT} , which covers the whole nodule rather than a rim, because the disagreement is about existence. Rows 2–3: four samples and the resulting U for the baseline and for our full model; both reproduce the ambiguity by emitting a mix of empty and non-empty samples. Row 4: the predicted $\hat{D}$ and the two error maps. $|\hat{D} - D^{GT}|$ is dark over the lesion while $|U - D^{GT}|$ is bright, which is the limitation discussed in Section 5: the head recovers existence-disagreement, but U would have to be maximal over the entire nodule to match D^{GT} , and that fights the segmentation loss.

- **Divergence and a runaway prior.** With the corrected (summed) loss every arm produced NaNs in the prior encoder within 2k steps, fixed with gradient clipping and a ± 10 clamp on the latent log-variance. And at $\lambda_A = 10^{-4}$ the KL doubled every step until overflow, because L_{align} back-propagates into the prior and inflating the prior variance is the cheapest way to raise diversity; fixed by detaching the prior in the sampling path and using $\lambda_A = 10^{-5}$.
- **Model selection.** Selecting the checkpoint on best validation GED picks step 1000–3000 for arms A and B — a barely-trained model with a wide, uninformative prior scores well on GED. We therefore report the matched 100k-step checkpoint for every arm, which compares methods rather than training length.
- **The conceptual limit (Figure ??).** On LIDC much of the disagreement is about lesion existence, so matching U to D^{GT} demands maximal uncertainty over *whole* lesions, which directly opposes the reconstruction term. This bounds how far L_{align} can go, and points at the fix in Section 6.

5 Conclusion

- **Q1 is confirmed:** where radiologists will disagree is predictable from the image alone — 0.627 correlation against a 0.425 trivial control, from a single 1×1 convolution. **Q2 is supported but modest:** the alignment loss raises $\text{corr}(U, D^{GT})$ by 0.026 and gives the best GED of our ELBO arms, at a 4% Dice cost.
- **The claim we make** is not “our Dice is higher”. The baseline already generates diverse plausible segmentations; our extension makes that diversity correspond to where experts actually disagree, and makes the disagreement itself an explicit, inspectable output.
- **One sentence for the whole design:** joint structure from the CVAE, marginal localisation from our loss. Arm D shows what happens when you try to get the joint from the loss instead — the best point estimate and no distribution at all ($E[d(S, S')] \approx 0$, 99–100% nested).

- **The sharpest limitation is the marginal itself.** D^{GT} is a per-pixel quantity: two graders drawing mask A and two drawing mask B produce exactly the same D^{GT} as four graders disagreeing incoherently. Neither D^{GT} nor U can tell coherent modes from pixel noise, which bounds both the supervision and the evaluation.
- **Other limitations:** 100k of the paper’s 240k iterations and crop-only augmentation, so our absolute GED sits above published values; one dataset and anatomy; D^{GT} estimated from only four graders; one seed per arm, so arm B’s Dice gap is not separable from run-to-run noise.
- **Future work:** restrict L_{align} to a boundary band so it shapes edge placement without demanding uncertainty over whole lesions — the most likely route to a strong Q2; rescue arm D by adding an explicit diversity/repulsion term or retaining the KL; and compare against the Hierarchical Probabilistic U-Net and MoSE on the same split.

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